

Data Preprocessing Summary

The following preprocessing steps were completed:

- Removed columns with more than 40% missing values.
- Filled missing numerical values using the median.
- Filled missing categorical values using the mode.
- Removed duplicate records.
- Detected and removed outliers using the IQR method.
- Encoded categorical variables using Label Encoding.
- Standardized numerical features using StandardScaler.
- Saved the cleaned dataset for exploratory data analysis and machine learning.